

Customer Churn Prediction Using Machine Learning

A project submitted for the Machine Learning
1 course (DS2401)

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ABSTRACT

This project aims to predict customer churn in a telecommunications company using machine learning techniques. The dataset was obtained from Kaggle and contains information about 7,043 customers described by 21 columns related to demographics, services, contracts, and billing information. The problem was defined as a binary classification task where the target variable is Churn. Data preprocessing included removing unnecessary attributes, handling missing values, converting data types, applying one-hot encoding to categorical variables, and scaling the features for Logistic Regression. Two machine learning models were implemented: Logistic Regression and Random Forest Classifier. Because the classes are imbalanced (about 27% of customers churn), the models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Logistic Regression achieved the highest accuracy (81.97%) and ROC-AUC (0.862), while Random Forest (accuracy 79.56%, ROC-AUC 0.859) with balanced class weights and a tuned decision threshold of 0.4 was selected as the final model because it detected 84% of the customers who churned. The results showed that customers with month-to-month contracts and shorter subscription periods were more likely to leave the company. The project demonstrates how machine learning can help organizations identify customers at risk of churn and improve customer retention strategies.

INTRODUCTION

Customer churn is one of the most important challenges faced by telecommunication companies. Churn occurs when customers stop using a company's services and move to competitors. Predicting customer churn allows companies to identify customers who are likely to leave and take preventive actions to improve customer retention.

METHODOLOGY

Data Source & Description:

The dataset was obtained from Kaggle (Telco Customer Churn Dataset). It contains 7,043 customer records and 21 attributes. The dataset includes both numerical and categorical variables describing customer demographics, account information, subscribed services, billing details, and customer churn status.

The target variable is Churn, which indicates whether a customer left the company (Yes) or remained with the company (No).

Data Preprocessing:

Several preprocessing steps were performed before model training:

1. The customerID column was removed because it does not contribute to churn prediction.
2. The TotalCharges column was converted from string format to numerical format.
3. Eleven missing values found in TotalCharges were replaced using the median value.
4. Categorical variables were transformed into numerical variables using One-Hot Encoding.
5. The target variable (Churn) was converted into binary values (0 and 1).

In addition, the features were standardized using StandardScaler (applied through a pipeline for Logistic Regression) so that the model converges properly. These steps ensured that the dataset was suitable for machine learning algorithms.

Algorithm/Model Selection:

Two machine learning models were selected:

1. Logistic Regression
2. Random Forest Classifier

Logistic Regression was chosen because it is a widely used classification algorithm that performs well on binary classification problems. Random Forest was selected because it is an ensemble learning method capable of capturing complex relationships within the data. Since only about 27% of the customers churn, the Random Forest was trained with balanced class weights (300 trees, minimum leaf size of 3) to give more importance to the churn class.

Training Procedure:

The dataset was divided into training and testing sets using an 80%-20% split. The training set was used to train the models, while the testing set was used to evaluate performance on unseen data.

Evaluation Strategy:

Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Accuracy alone can be misleading for imbalanced data (predicting that no customer leaves would already give about 73% accuracy), so recall and F1-score for the churn class and ROC-AUC were used as the main comparison metrics. The decision threshold was also tuned (0.3 to 0.6) to balance precision and recall.

Libraries and functions:

The project was implemented using Python and the following libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

RESULTS AND DISCUSSION

The exploratory analysis revealed several important patterns in customer churn behavior.

First, approximately 73.46% of customers remained with the company, while 26.54% left the service.

Second, contract type showed a strong relationship with customer churn. Customers with month-to-month contracts had the highest churn rate, while customers with one-year and two-year contracts were significantly less likely to leave.

Third, customer tenure was strongly associated with churn. Customers with shorter subscription periods were more likely to leave the company, whereas long-term customers showed greater loyalty.

Two classification models were trained and evaluated. Precision, recall and F1-score in the table refer to the churn class.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	81.97%	0.68	0.60	0.64	0.862
Random Forest (threshold 0.5)	79.56%	0.60	0.71	0.65	0.859
Random Forest (threshold 0.4) - final	**76.58%**	**0.54**	**0.84**	**0.66**	**0.859**

Logistic Regression achieved the highest accuracy (81.97%) and the highest ROC-AUC (0.862), but it detected only 60% of the customers who actually churned (recall = 0.60). The Random Forest model, trained with balanced class weights, detected more churners (recall = 0.71) with a similar ROC-AUC (0.859). This shows that both models rank customers by churn risk almost equally well, and that the main difference comes from the decision threshold.

Lowering the Random Forest decision threshold from 0.5 to 0.4 increased recall to 0.84 and gave the best F1-score for the churn class (0.66), at the cost of lower precision (0.54) and lower accuracy (76.58%). Since missing a customer who is about to leave is usually more costly than

offering a retention incentive to a loyal customer, Random Forest with a threshold of 0.4 was selected as the final model.

The confusion matrix of the final model (Figure 1) shows that 315 of the 373 customers who churned were correctly identified (58 were missed), while 272 of the 1,036 loyal customers were flagged as potential churners.

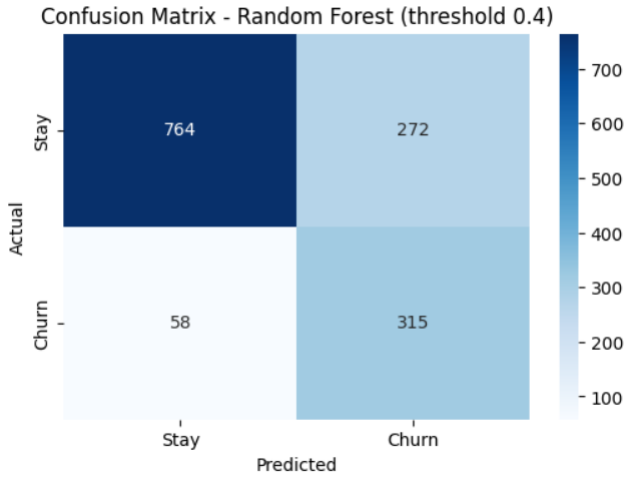


Figure 1: Confusion matrix of the final Random Forest model (threshold 0.4).

The Random Forest feature importance showed that tenure, TotalCharges and MonthlyCharges were the most influential variables, followed by contract type (two-year and one-year), fiber optic internet service, electronic check payment, online security and tech support. These results are consistent with the exploratory analysis, in which new customers and month-to-month contracts showed the highest churn.

One limitation of the project is that the results come from a single train/test split, and cross-validation and a systematic hyperparameter search were not applied; only the decision threshold was tuned. Future work could improve performance by exploring more advanced algorithms (such as gradient boosting) and feature engineering methods.

CONCLUSION

This project successfully applied machine learning techniques to predict customer churn using the Telco Customer Churn dataset. After preprocessing the data and training two classification models, Logistic Regression achieved the highest accuracy (81.97%) and ROC-AUC (0.862), while Random Forest with balanced class weights and a decision threshold of 0.4 was selected as the final model because it detected 84% of the customers who churned (recall = 0.84, F1-score = 0.66).

The analysis showed that tenure, monthly and total charges, and contract type were among the most influential factors affecting customer churn. Customers with month-to-month contracts and shorter subscription periods were more likely to leave the company.

Although the models produced good results, future improvements may include cross-validation, hyperparameter tuning, and testing additional machine learning algorithms. The findings demonstrate the practical value of machine learning in helping companies identify at-risk customers and improve retention strategies, for example by offering incentives for longer contracts to new customers.

REFERENCES

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THE EXPERIENCES AND SKILLS ACQUIRED BY THE STUDENT:

Student Name	**Experiences**	**Skills**
Rand alhasnani	Handled data preprocessing and cleaning, Performed exploratory data analysis and visualization, Prepared the report and interpreted the findings, Implemented Random Forest model and compared results, Implemented Logistic Regression model, Handled class imbalance and tuned the decision threshold, Evaluated models using precision, recall, F1-score and ROC-AUC.	Data cleaning, handling missing values, feature transformation and scaling, Data visualization, interpretation of charts, Machine learning model development, Model evaluation and comparison (precision, recall, F1-score, ROC-AUC), Threshold tuning for imbalanced data